ANALYSIS IV — CHEAT SHEET (EPFL) · Complex analysis · Laplace · Fourier · PDEs

0. ℂ TOOLBOX + EXAM STRATEGY

$z = x + iy = re^{i\theta}$, $r = |z| = \sqrt{x^2+y^2}$, $\theta = \arg z$, $x = r\cos\theta$, $y = r\sin\theta$.

$e^{i\theta} = \cos\theta + i\sin\theta$, $\bar{z} = x - iy$, $|z|^2 = z\bar{z}$, $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$.

De Moivre: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
Principal arg $\operatorname{Arg} z \in (-\pi, \pi]$: $\arctan\frac{y}{x}$ if $x>0$; $\pm\pi + \arctan\frac{y}{x}$ if $x<0$ ($+$ if $y\ge 0$).
Key values: $e^{i\pi/2}=i$, $e^{i\pi}=-1$, $e^{-i\pi/2}=-i$, $e^{2ik\pi}=1$.
n-th roots of $w = \rho e^{i\varphi}$: $z_k = \rho^{1/n} e^{i(\varphi+2k\pi)/n}$, $k=0..n-1$.
Identities: $\sin^2+\cos^2=1$, $\cosh^2-\sinh^2=1$, $\sin 2\theta = 2\sin\theta\cos\theta$, $\cos 2\theta = 1-2\sin^2\theta$, $\cos(iz)=\cosh z$, $\sin(iz)=i\sinh z$, $|e^z|=e^{\operatorname{Re} z}$.

> △ **Strategy (negative marking!)** Wrong MCQ $= -\frac{1}{3}$ (or -1); wrong True/False $= -1$. **Answer only if sure**; total guess → leave blank (0). Open questions: *justify every step* (points follow steps).
> **Index:** $\oint / \int_\gamma$ →§5 | $\int_0^{2\pi}$ →§6 | ODE+IC→§7 | PDE on $[0,L]$ →§10a | PDE on $\mathbb{R}$ →§10b. Laplace/Fourier tables are *given* at the exam.

1. ELEMENTARY FNS + HOLOMORPHY (CR)

$e^z = e^x(\cos y + i\sin y)$, $|e^z| = e^x$, period $2\pi i$.

$$\cos z = \frac{e^{iz}+e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz}-e^{-iz}}{2i},$$
$$\cosh z = \frac{e^z+e^{-z}}{2}, \qquad \sinh z = \frac{e^z-e^{-z}}{2}.$$

$\operatorname{Log} z = \ln|z| + i\operatorname{Arg} z$ (cut on $\mathbb{R}_{\le 0}$); $z^a = e^{a\operatorname{Log} z}$.
Cauchy–Riemann: $f = u + iv$ holomorphic $\iff$ $u, v \in C^1$ *and*

$\boxed{u_x = v_y, \quad u_y = -v_x}$, $\quad f'(z) = u_x + iv_x = u_x - iu_y$.

Harmonic: u holo. part $\Rightarrow \Delta u = u_{xx} + u_{yy} = 0$.

1. **Test:** both CR equations must hold everywhere.
2. **Harmonic conjugate** v **from** u: integrate $v = \int u_x\,dy + g(x)$, then impose $v_x = -u_y$ to get $g'(x)$ (v unique up to a real constant).
3. f': compute $u_x - iu_y$, then set $y=0, x=z$ to read f in z.

> **Ex.** *(24/Q24)* $u = e^{x^2-y^2}\cos 2xy \Rightarrow f' = 2z\,e^{z^2}$, $f = e^{z^2}$.
> **Ex.** *(25)* $u = x^3 - 3xy^2 + 1 \Rightarrow v = 3x^2y - y^3 + C$.
> **Ex.** $f = x + ay + i(bx+cy)$ holo. $\iff$ $c=1$, $a=-b$.

> △ $\operatorname{Re} z, \operatorname{Im} z, \bar{z}, |z|$ are **never** holomorphic. $1/z$: holo. on $\mathbb{C}\setminus\{0\}$, not on all $\mathbb{C}$. Every entire fn (and its derivative) satisfies CR everywhere.

> **True/False quick wins:** $\oint$ of an entire fn over a closed curve $= 0$ **(T)**; $e^{1/z}$ has an order-1 pole at 0 **(F: essential)**; $1/z$ holo. on all $\mathbb{C}$ **(F)**; deriv. of $z\cos z$ satisfies CR **(T)**; $\operatorname{Re}(z)$ holomorphic **(F)**.

2. SINGULARITIES: TYPE & ORDER

See: "poles?", "order", "2nd-order pole?", essential.

1. **Factor** N and D ($z^2-1=(z-1)(z+1)$, $z^2+1=(z-i)(z+i)$, $z^3-z = z(z-1)(z+1)$).
2. **Cancel** common factors: each cancelled $(z-a) \Rightarrow$ **removable** (NOT a pole) — *trap #1.*
3. Order of pole at a = (zero order of D) − (zero order of N), if >0. Else removable ($=0$) or regular (<0).
4. $\sin g = 0 \iff g = k\pi$; $\cos g = 0 \iff g = \frac{\pi}{2} + k\pi$ (simple zeros).
5. **Essential:** $e^{1/z}, e^{1/z^2}, \sin\frac{1}{z}$ (∞ negative powers) — never a pole.

> **Ex.** $\frac{z^2-4}{(z-2)(z+3)^2} = \frac{z+2}{(z+3)^2}$: $z=2$ removable, $z=-3$ **order 2**.
> **Ex.** $\frac{z^2+z}{(z^2-1)^2} = \frac{z}{(z-1)^2(z+1)}$: order 2 at 1, order 1 at -1.
> **Ex.** $\frac{z^2-1}{\sin(z+1)}$: poles $z = -1 + k\pi$, $k \in \mathbb{Z}\setminus\{0\}$ ($k=0$ removable).
> **Ex.** $e^{1/z}$ at 0: essential ⇒ "order-1 pole" = **FALSE**.

> **Classify** z_0: *removable* ⇔ $\lim f$ finite (no neg. powers); *pole order* m ⇔ $|f|\to\infty$ & $(z-z_0)^m f$ holo. $\neq 0$ (finitely many neg. powers); *essential* ⇔ $\lim$ DNE (∞ neg. powers; image dense, Casorati–Weierstrass).

3. TAYLOR & LAURENT / SINGULAR PART

Taylor (f holo. in disk): $f = \sum_{n\ge 0} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$; radius = distance to nearest singularity.

Laurent (annulus $r<|z-z_0|<R$ between singularities): $f = \sum_{n=-\infty}^{\infty} c_n (z-z_0)^n$, unique.
Singular (principal) part of $f = \frac{g(z)}{(z-z_0)^k}$ (g holo., $g(z_0)\neq 0$): expand $g = \sum a_m (z-z_0)^m$, keep powers <0:

$$\sum_{m=0}^{k-1} \frac{a_m}{(z-z_0)^{k-m}} \qquad (\text{need } a_0, \dots, a_{k-1}).$$

Standard Maclaurin series (need them by heart):

$$e^z = \sum_{n\ge 0}\frac{z^n}{n!}, \quad \frac{1}{1-z} = \sum_{n\ge 0} z^n, \quad \frac{1}{1+z} = \sum (-1)^n z^n,$$
$$\sin z = \sum \frac{(-1)^n z^{2n+1}}{(2n+1)!}, \quad \cos z = \sum \frac{(-1)^n z^{2n}}{(2n)!},$$
$$\sinh z = \sum \frac{z^{2n+1}}{(2n+1)!}, \quad \cosh z = \sum \frac{z^{2n}}{(2n)!},$$
$$\frac{1}{(1-z)^2} = \sum (n+1) z^n, \quad \operatorname{Log}(1+z) = \sum_{n\ge 1} \frac{(-1)^{n+1}}{n} z^n.$$

cos, cosh even; sin, sinh odd. Geometric/Log valid $|z|<1$.
Parity: f even ⇒ only even powers ⇒ $c_{\text{odd}}=0$; f odd ⇒ $c_{\text{even}}=0$.

> **Ex.** $\frac{z+1}{z^3(z^2+1)}$ at 0: $g = 1 + z - z^2 + \cdots \Rightarrow \frac{1}{z^3} + \frac{1}{z^2} - \frac{1}{z}$.
> **Ex.** $\frac{e^{z-i}}{z^2+1}$ at i: simple pole, $\frac{-i}{2(z-i)}$.
> **Ex.** $c_{2n}=0$ (odd): $z^2\sinh z$. $c_{2n+1}=0$ (even): $e^{1/z^2}\cosh z$.

> △ "Singular part" = **negative** powers only. "$c_{2n}=0$" ⇒ **odd** function (counter-intuitive).

4. RESIDUES

$\operatorname{Res}_{z_0} f = c_{-1}$ (coeff. of $\frac{1}{z-z_0}$).

$$\text{simple: } \operatorname{Res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f = \frac{p(z_0)}{q'(z_0)}.$$

$$\text{order } m: \operatorname{Res}_{z_0} f = \frac{1}{(m-1)!}\lim_{z\to z_0}\frac{d^{m-1}}{dz^{m-1}}\left[(z-z_0)^m f\right].$$

($m=2$: one derivative, $\frac{1}{(m-1)!}=1$.) **Essential:** read c_{-1} from Laurent.

> **Ex.** $\frac{z^3}{(z-1)^2(z-2)}$ at 1: $\frac{d}{dz}\frac{z^3}{z-2}\big|_1 = -4$.
> **Ex.** $\frac{e^z}{(z-1)^2(z-2)}$ at 1: $\frac{d}{dz}\frac{e^z}{z-2}\big|_1 = -2e$.
> **Ex.** Residue 0 at 0: e^{1/z^2} (only z^{-2k}). But $\frac{1}{z}, \frac{\sin z}{z^2}, e^{1/z}$ have $c_{-1}=1$.

> △ Residue is a *number* (-4); integral $= 2\pi i \cdot$ (sum).

5. CONTOUR INTEGRATION

> **Cauchy thm:** f holo. on simply-conn. ⇒ $\oint_\gamma f = 0$, path-independent.
> **Cauchy integral formula:** z_0 inside γ (CCW), f holo.:
> $$f(z_0) = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{z-z_0}dz,$$
> $$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_\gamma \frac{f(z)}{(z-z_0)^{n+1}}dz.$$
> **Residue thm:** $\oint_\gamma f = 2\pi i \sum_{z_k \text{ inside}} \operatorname{Res}_{z_k} f$.
> **ML bound:** $|\int_\gamma f| \le (\max_\gamma |f|)\cdot \text{length}(\gamma)$.
> **Liouville:** bounded & entire ⇒ constant. **Fund. thm. algebra:** nonconstant poly has a root in $\mathbb{C}$. **Max modulus:** nonconstant holo. ⇒ $|f|$ has no interior max.

(A) Open path $z_0 \to z_1$, f holo.: $\int_\gamma f = F(z_1) - F(z_0)$ ($F'=f$). $\int_\gamma \frac{1}{z} = \operatorname{Log} z_1 - \operatorname{Log} z_0$ (if it avoids $\mathbb{R}_{<0}$).
(B) Closed: residue thm (keep only *interior* singularities; f entire ⇒ 0).
(C) "Possible values" over arbitrary closed γ:

$$\{2\pi i \sum_{k\in S}\operatorname{Res}_{z_k} f : S \subseteq \{\text{sing.}\}\}, \text{ incl. } \varnothing \to 0.$$

(D) Two paths, same ends: $\int_{\gamma_1} \neq \int_{\gamma_2} \iff$ a singularity with *nonzero residue* lies between them.

> **Ex.** $\oint_{|z|=1}\frac{z}{z-a} = 2\pi i\,a$ if $|a|<1$; $= \pi i \Rightarrow a = \frac{1}{2}$.
> **Ex.** $\oint_{|z|=2}\frac{e^{\pi z/2}}{z^2+1}$: $\operatorname{Res}_{\pm i} = \frac{1}{2} \Rightarrow 2\pi i$.
> **Ex.** $\int z^2$, $1\to i$: $\frac{-1-i}{3}$. $\int \frac{1}{z}$, $i \to 1$: $-\frac{i\pi}{2}$.
> **Ex.** *(Q25)* $\frac{\sin z^2}{z^3(z-2i)}$: values $\{0, -\pi, \frac{\pi\sin 4}{4}, -\pi + \frac{\pi \sin 4}{4}\}$.
> **Ex.** *(Q1)* $\frac{\cos z}{z(z-\pi/2)^2}$: values $\{0, \frac{8i}{\pi}, -4i, \frac{8i}{\pi} - 4i\}$.

6. REAL INTEGRALS VIA RESIDUES

(A) $\int_0^{2\pi} R(\cos t, \sin t)\,dt$: $z = e^{it}$ ($|z|=1$): $\cos t = \frac{z^2+1}{2z}$, $\sin t = \frac{z^2-1}{2iz}$, $dt = \frac{dz}{iz}$.

$$\int_0^{2\pi} R\,dt = 2\pi i \sum_{|z_k|<1}\operatorname{Res}_{z_k}\left[R\,\frac{1}{iz}\right].$$

Keep poles with $|z|<1$; result must be **real**.

> **Ex.** $\int_0^{2\pi}(2+\cos t)^2 dt = \frac{1}{4i}\oint\frac{(z^2+4z+1)^2}{z^3}$; res at 0 (ord. 3) = coeff. $z^2 = 18 \Rightarrow$ **9π**.
> **Ex.** $\int_0^{2\pi}\frac{5/4-\sin t}{5/4+\cos t}dt$: poles $0, -\frac{1}{2}$; $\operatorname{Res}_0 = -1$, $\operatorname{Res}_{-1/2} = 1 - \frac{5i}{3}$; ⇒ **$\frac{10\pi}{3}$**.

(B) $\int_{-\infty}^{\infty}$ — close a big semicircle in the *upper* half-plane. Rational R with $\deg D \ge \deg N + 2$:

$$\int_{-\infty}^{\infty} R\,dx = 2\pi i \sum_{\operatorname{Im} z_k > 0}\operatorname{Res}_{z_k} R.$$

Jordan ($a>0$): $\int_{-\infty}^{\infty} f(x)e^{iax}dx = 2\pi i \sum_{\operatorname{Im} z_k>0}\operatorname{Res}_{z_k}[f e^{iaz}]$; take Re / Im for $\cos ax$ / $\sin ax$.

> **Ex.** $\int_{-\infty}^{\infty}\frac{dx}{x^2+1} = 2\pi i\,\operatorname{Res}_i = 2\pi i \cdot \frac{1}{2i} = \pi$.

7. LAPLACE TRANSFORM

$\mathcal{L}[f](z) = \int_0^\infty f(t)e^{-zt}dt$, holo. for $\operatorname{Re} z > \gamma_0$ (abscissa).
Table ($\mathcal{L}[f] = F$):
$1 \to \frac{1}{z}$; $t^n \to \frac{n!}{z^{n+1}}$; $e^{at} \to \frac{1}{z-a}$; $te^{at} \to \frac{1}{(z-a)^2}$; $\sin\omega t \to \frac{\omega}{z^2+\omega^2}$; $\cos\omega t \to \frac{z}{z^2+\omega^2}$; $\sinh\omega t \to \frac{\omega}{z^2-\omega^2}$; $\cosh\omega t \to \frac{z}{z^2-\omega^2}$; $e^{at}\sin\omega t \to \frac{\omega}{(z-a)^2+\omega^2}$; $e^{at}\cos\omega t \to \frac{z-a}{(z-a)^2+\omega^2}$; $t\cos\omega t \to \frac{z^2-\omega^2}{(z^2+\omega^2)^2}$. ***Properties:***

$$\mathcal{L}[y'] = zY - y(0), \quad \mathcal{L}[y''] = z^2Y - zy(0) - y'(0),$$
$$\mathcal{L}[t^n f] = (-1)^n F^{(n)}, \quad \mathcal{L}[e^{at}f] = F(z-a),$$
$$\mathcal{L}[\textstyle\int_0^t f] = \frac{F}{z}, \quad \mathcal{L}^{-1}[F\,G] = (f*g)(t) = \int_0^t f(t-s)g(s)ds.$$

Partial fractions (to invert):
$\frac{P(z)}{Q(z)}$ with $Q = \prod (z-r_k)^{m_k}$: write $\sum_k \sum_{j=1}^{m_k}\frac{A_{k,j}}{(z-r_k)^j}$. Simple root: $A_k = \frac{P(r_k)}{Q'(r_k)}$ (cover-up). Irreducible z^2+pz+q: keep $\frac{Bz+C}{z^2+pz+q}$, complete the square. ***ODE recipe:***

1. Apply $\mathcal{L}$ (insert IC), isolate $Y = \frac{\text{IC}+\mathcal{L}[\text{rhs}]}{z^2+bz+a}$.
2. Invert: partial fractions, or **complete the square** $z^2+2pz+a = (z+p)^2+\omega^2$, $\omega^2 = a - p^2$.
3. $\mathcal{L}^{-1}\frac{\omega}{(z+p)^2+\omega^2} = e^{-pt}\sin\omega t$; $\frac{z+p}{(\cdot)} = e^{-pt}\cos\omega t$; $\frac{1}{(z+p)^2} = te^{-pt}$.
4. Check $y(0)$ *and* $y'(0)$.

> **Ex.** $y''+2y'+y=t$, $y(0)=0, y'(0)=3$ ⇒ $y = t - 2 + 2e^{-t} + 4te^{-t}$.
> **Ex.** $\mathcal{L}[t\cosh 4t] = \frac{z^2+16}{(z^2-16)^2}$; $\mathcal{L}^{-1}\frac{z}{(z+1)^2} = (1-t)e^{-t}$.

> △ cos: $z^2+\omega^2$ but cosh: $z^2-\omega^2$. Numerator of sin / sinh is ω, of cos / cosh is z.

8. FOURIER SERIES (period T, $\omega_0 = \frac{2\pi}{T}$)

$$f(x) = \frac{a_0}{2} + \sum_{n\ge 1}[a_n\cos n\omega_0 x + b_n \sin n\omega_0 x]$$
$$= \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 x} \quad (\text{complex form}).$$
$$a_n = \frac{2}{T}\int_T f\cos n\omega_0 x\,dx, \quad b_n = \frac{2}{T}\int_T f\sin n\omega_0 x\,dx,$$
$$c_n = \frac{1}{T}\int_T f\,e^{-in\omega_0 x}dx.$$

$c_0 = \frac{a_0}{2}$, $c_n = \frac{a_n - ib_n}{2}$ ($n>0$). f **even**⇒ $b_n=0$ (cosines); **odd**⇒ $a_n=0$ (sines).
Half-range on $[0,L]$ (used in PDEs): sine $f = \sum b_n \sin\frac{n\pi x}{L}$, $b_n = \frac{2}{L}\int_0^L f\sin\frac{n\pi x}{L}dx$; cosine $f = \frac{a_0}{2} + \sum a_n\cos\frac{n\pi x}{L}$, $a_n = \frac{2}{L}\int_0^L f\cos\frac{n\pi x}{L}dx$.
Parseval: $\frac{1}{T}\int_T |f|^2 = \sum |c_n|^2 = \frac{a_0^2}{4} + \frac{1}{2}\sum(a_n^2 + b_n^2)$.
At a jump the series → $\frac{f(x^+)+f(x^-)}{2}$.

> **Ex.** Constant 1 on $(0,1)$: $b_n = \frac{2(1-(-1)^n)}{n\pi}$ ($\frac{4}{n\pi}$ if n odd, else 0).

9. FOURIER TRANSFORM

$$\hat{f}(\alpha) = \mathcal{F}[f] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)e^{-i\alpha x}dx,$$
$$f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\hat{f}(\alpha)e^{i\alpha x}d\alpha \quad (\text{inverse}).$$

Properties:

$$\mathcal{F}[f^{(n)}] = (i\alpha)^n\hat{f}, \quad \mathcal{F}[x^n f] = i^n \hat{f}^{(n)},$$
$$\mathcal{F}[f(x-a)] = e^{-i\alpha a}\hat{f}, \quad \mathcal{F}[e^{iax}f] = \hat{f}(\alpha - a),$$
$$\mathcal{F}[f(\lambda x)] = \frac{1}{|\lambda|}\hat{f}\left(\frac{\alpha}{\lambda}\right), \quad \mathcal{F}[f*g] = \sqrt{2\pi}\,\hat{f}\hat{g}.$$

Plancherel: $\int|f|^2=\int|\hat f|^2$. **Gaussian:** $e^{-x^2/2}\to e^{-\alpha^2/2}$; $e^{-ax^2}\to\frac{1}{\sqrt{2a}}e^{-\alpha^2/4a}$.

Pairs: $\frac{1}{x^2+\omega^2}\to\sqrt{\frac{\pi}{2}}\,\frac{e^{-\omega|\alpha|}}{\omega}$; $\mathbf{1}_{|x|<b}\to\sqrt{\frac{2}{\pi}}\,\frac{\sin b\alpha}{\alpha}$; $e^{-a|x|}\to\sqrt{\frac{2}{\pi}}\,\frac{a}{a^2+\alpha^2}$; $e^{-\omega x}\mathbf{1}_{x>0}\to\frac{1}{\sqrt{2\pi}(\omega+i\alpha)}$ $(\omega>0)$; $\mathbf{1}_{b<x<c}\to\frac{e^{-ib\alpha}-e^{-ic\alpha}}{\sqrt{2\pi}\,i\alpha}$.

△ $(i\alpha)^2=-\alpha^2$, $(i\alpha)^4=+\alpha^4$ $(i^0,i^1,i^2,i^3=+,+,-,-)$. f real & even $\Rightarrow$ $\hat f$ real.

10. PDEs — WHICH TRANSFORM?

Rule by spatial domain: $x\in[0,L]$ Dirichlet $\to$ Fourier *sine* series (§10a); $x\in\mathbb{R}\to$ Fourier transform (§10b); $x\in[0,\infty)\to$ Laplace in x.

Separation of variables: try $u=X(x)T(t)$; spatial part solves $X''+\lambda X=0$ with the BC. Dirichlet $X(0)=X(L)=0\Rightarrow\lambda_n=(\frac{n\pi}{L})^2$, $X_n=\sin\frac{n\pi x}{L}$; Neumann $X'(0)=X'(L)=0\Rightarrow X_n=\cos\frac{n\pi x}{L}$. Superpose, match IC by Fourier coeff.

(10a) Bounded $[0,L]$**, Dirichlet:** $u=\sum b_n(t)\sin\frac{n\pi x}{L}$; $u_{xx}\to-(\frac{n\pi}{L})^2b_n$, $u_{xxxx}\to+(\frac{n\pi}{L})^4b_n$. Get ODE in b_n; set $b_n(0),b_n'(0)$ by projecting IC $(\frac{2}{L}\int_0^L\cdot\sin)$. If IC $=\sin(k\pi x/L)$: only $n=k$.

Ex. *(Schröd.)* $iu_t=u_{xx}-u$: $b_n'=i((n\pi)^2+1)b_n$ $(\frac{1}{i}=-i)$.

(10b) Unbounded $\mathbb{R}$**:** apply $\mathcal{F}$ in x, t = parameter; PDE $\to$ ODE in $\hat u(\alpha,t)$:

$$\hat u_t+a(t)\hat u=0\;\Rightarrow\;\hat u=\hat g\,e^{-\int_0^t a(s)ds},$$
$$\hat u_{tt}+\omega^2\hat u=0\;\Rightarrow\;\hat u=\hat u_0\cos\omega t+\frac{\hat v_0}{\omega}\sin\omega t.$$

(+ sign $\to$ cos / sin, − sign $\to$ cosh / sinh.)

Wave on $\mathbb{R}$ $u_{tt}=c^2u_{xx}$ (d'Alembert):

$$u=\frac{u_0(x+ct)+u_0(x-ct)}{2}+\frac{1}{2c}\int_{x-ct}^{x+ct}u_1(s)ds.$$

Heat on $\mathbb{R}$ $u_t=u_{xx}$: $u=\int G(x-y,t)u_0(y)dy$, $G(x,t)=\frac{1}{\sqrt{4\pi t}}e^{-x^2/4t}$.

Ex. *(24)* $u_t+(2t+1)u_{xxxx}=0$: $\hat u=\hat g\,e^{-\alpha^4(t^2+t)}$.
Ex. *(25)* $u_{tt}=u_{xx}-u$, $\omega=\sqrt{\alpha^2+1}$: $\hat u=\hat u_0\cos(\omega t)+\frac{\hat v_0}{\omega}\sin(\omega t)$.

△ $u_{xxxx}\to+(\frac{n\pi}{L})^4$ (no minus). $u_{tt}+2u_t\Rightarrow e^{-t}$; $u_{tt}-2u_t\Rightarrow e^{+t}$. t-coeff. on $u_{xx}\Rightarrow t^2$ in exp.

WORKED SOLUTIONS — OPEN QUESTIONS (big points)

These mirror the recurring open questions: copy the ***structure****, change the numbers. Justify each step.*

▶ A. $f'(z)$ via Cauchy–Riemann (≈8 pts)

24/Q24. $u=e^{x^2-y^2}\cos 2xy$.

1. $u_x=e^{x^2-y^2}(2x\cos 2xy-2y\sin 2xy)$, $u_y=e^{x^2-y^2}(-2y\cos 2xy-2x\sin 2xy)$.
2. $f'=u_x-iu_y=2(x+iy)e^{x^2-y^2}(\cos 2xy+i\sin 2xy)=2z\,e^{(x+iy)^2}$.

$\boxed{f'(z)=2z\,e^{z^2}}$ (so $f=e^{z^2}$).

▶ B. Possible values of $\oint_\gamma f$ (≈9–10 pts)

Find singularities, each residue, then $\{2\pi i\sum_S\text{Res}:S\subseteq\text{sing.}\}$, $\varnothing\to 0$.

(Q25) $f=\frac{\sin z^2}{z^3(z-2i)}$: $\frac{\sin z^2}{z^3}=\frac{1}{z}-\frac{z^3}{6}+\dots$
$\Rightarrow\text{Res}_0=-\frac{1}{2i}=\frac{i}{2}$; $\text{Res}_{2i}=\frac{\sin(-4)}{-8i}=-\frac{i\sin 4}{8}$.
$\Rightarrow\boxed{\{0,\ -\pi,\ \frac{\pi\sin 4}{4},\ -\pi+\frac{\pi\sin 4}{4}\}}$.

(Q1) $f=\frac{\cos z}{z(z-\pi/2)^2}$: $\text{Res}_0=\frac{4}{\pi^2}$;
$\text{Res}_{\pi/2}=\frac{d}{dz}\frac{\cos z}{z}\Big|_{\pi/2}=-\frac{2}{\pi}$.
$\Rightarrow\boxed{\{0,\ \frac{8i}{\pi},\ -4i,\ \frac{8i}{\pi}-4i\}}$.

▶ C. $\int_0^{2\pi}R(\cos t,\sin t)\,dt$ (≈10–12 pts)

(Q26) $(2+\cos t)^2$: $I=\frac{1}{4i}\oint\frac{(z^2+4z+1)^2}{z^3}$; res = coeff. $z^2=18\Rightarrow\boxed{9\pi}$.

(Q3) $\frac{5/4-\sin t}{5/4+\cos t}$ $\to$ $f=\frac{4z^2-10iz-4}{2z(2z+1)(z+2)}$; poles $0,-\frac{1}{2}$;
$\text{Res}_0=-1$, $\text{Res}_{-1/2}=1-\frac{5i}{3}$; sum $-\frac{5i}{3}\Rightarrow\boxed{\frac{10\pi}{3}}$.

▶ D. ODE via Laplace (≈11–12 pts)

(Q27) $y''+2y'+y=t$, $y(0)=0,y'(0)=3$: $(z+1)^2Y=3+\frac{1}{z^2}$; $\frac{1}{z^2(z+1)^2}=-\frac{2}{z}+\frac{1}{z^2}+\frac{2}{z+1}+\frac{1}{(z+1)^2}$
$\Rightarrow\boxed{y=t-2+2e^{-t}+4te^{-t}}$.

(Q2a) $y''+2y'+ay=0$, $a>1$, $y'(0)=y_1$:
$\boxed{y=\frac{y_1}{\sqrt{a-1}}e^{-t}\sin(\sqrt{a-1}\,t)}$.

▶ E. Bounded PDE → Fourier sine (≈12–16 pts)

$u=\sum b_n\sin(n\pi x)$; $u_{xx}\to-(n\pi)^2b_n$, $u_{xxxx}\to+(n\pi)^4b_n$.

(Q28) $u_{tt}-2u_t=u_{xx}$, $u(x,0)=0$, $u_t(x,0)=1$: $b_n''-2b_n'+(n\pi)^2b_n=0$; $b_n(0)=0$, $b_n'(0)=c_n=\frac{4}{n\pi}$ (n odd); $r=1\pm i\omega_n$, $\omega_n=\sqrt{(n\pi)^2-1}$.

$$\boxed{u=\sum_{n\text{ odd}}\frac{4}{n\pi\omega_n}e^t\sin(\omega_n t)\sin(n\pi x)}.$$

(Q2b) $u_{tt}+2u_t=-u_{xxxx}-2u$, $u_t(x,0)=\sin 2\pi x$: only $n=2$, $b_2''+2b_2'+(16\pi^4+2)b_2=0$, $\omega=\sqrt{16\pi^4+1}$

$$\Rightarrow\boxed{u=\frac{e^{-t}}{\omega}\sin(\omega t)\sin 2\pi x}.$$

STEP-BY-STEP METHODS (open questions)

Follow the steps mechanically — they fit every variant seen on the past exams.

▷ Recover $f(z)$ from $\text{Re}\,f=u$ (or $\text{Im}\,f=v$):

1. Check u harmonic ($u_{xx}+u_{yy}=0$) — else no holomorphic f.
2. $v_y=u_x\Rightarrow v=\int u_x\,dy+g(x)$.
3. $v_x=-u_y\Rightarrow$ solve $g'(x)$, then integrate for g.
4. $f=u+iv$; set $y=0$ and replace $x\to z$. Check $f'=u_x-iu_y$.

▷ Possible values of $\oint_\gamma f$:

1. List all singularities S; compute Res at each.
2. Values $=\{2\pi i\sum_{s\in T}\text{Res}_s : T\subseteq S\}$; empty $T\Rightarrow 0$.
3. If γ may wind k times, weight each residue by its winding number.

▷ Classify a singularity z_0 + its order:

1. Factor, cancel common $(z-z_0)$ *first* (else a false pole).
2. Removable if lim finite; pole of order $m=\text{ord}_{z_0}D-\text{ord}_{z_0}N$.
3. Essential if ∞ negative Laurent powers ($e^{1/z}$, $\sin\frac{1}{z}$, …).

▷ $\int_0^{2\pi}R(\cos t,\sin t)\,dt$:

1. $z=e^{it}$: $\cos t=\frac{z^2+1}{2z}$, $\sin t=\frac{z^2-1}{2iz}$, $dt=\frac{dz}{iz}$.
2. Reduce to $\oint_{|z|=1}$ of a rational function; keep only poles $|z|<1$.
3. $I=2\pi i\sum\text{Res}$; the answer must come out *real*.

▷ ODE via Laplace:

1. Apply $\mathcal{L}$: $\mathcal{L}[y']=zY-y(0)$, $\mathcal{L}[y'']=z^2Y-zy(0)-y'(0)$.
2. Solve algebraically for $Y(z)$.
3. Partial fractions (complete the square for complex roots).
4. Invert term-by-term with the table.

▷ Bounded PDE on $[0,L]\to$ Fourier sine:

1. $u=\sum b_n(t)\sin\frac{n\pi x}{L}$ (Dirichlet; use cos for Neumann).
2. Substitute $u_{xx}\to-(\frac{n\pi}{L})^2b_n$, $u_{xxxx}\to+(\frac{n\pi}{L})^4b_n$: an ODE in t for each b_n.
3. $b_n(0),b_n'(0)$ by projecting the $t=0$ data: $b_n(0)=\frac{2}{L}\int_0^L u(x,0)\sin\frac{n\pi x}{L}\,dx$.
4. A single $\sin(k\pi x)$ datum excites only mode $n=k$.

PAST-EXAM ANSWER KEY (pattern bank)

Exact answers from the two exams you sent — the new exam reshuffles the same patterns.

2024: SCQ1 $c=1,a=-b$; SCQ2 $z=1,-2i$; SCQ3 $z=-1+k\pi$ $(k\neq 0)$; SCQ4 $z^2\sinh z$; SCQ5 $\frac{-i}{2(z-i)}$; SCQ6 $\frac{1}{z^3}+\frac{1}{z^2}-\frac{1}{z}$; SCQ7 $\int_{\Gamma_1}=\int_{\Gamma_2}$; SCQ9 $\frac{1}{z-1/2}$; SCQ10 $\frac{-1-i}{3}$; SCQ11 $2\pi i$; SCQ12 -4; SCQ13 $\frac{2z-1}{z^4-2z^3+z^2}$; SCQ14 $4\pi i$; SCQ15 $\frac{z}{z^2+4}$; SCQ16 $1-e^{-t}$; SCQ17 $F''(z)$; SCQ18 $\frac{i}{2\pi}\int_\gamma g$; SCQ19 $\hat g\,e^{-\alpha^4(t^2+t)}$; **T/F** 01 T, 04 F, 06 F, 08 T, 09 F.

2025: SCQ10 $3x^2y-y^3+2$; SCQ11 $a=\frac{1}{2}$; SCQ12 $\frac{e^z}{(z-1/2)^2+1/4}$; SCQ13 $-\frac{\pi i}{2}$; SCQ14 $e^{1/z^2}\cosh z$; SCQ15 $\frac{1}{z^3}+\frac{1}{z^2}-\frac{1}{2z}$; SCQ16 $\frac{1}{(z-1)^2(z^2-2z+2)}$; SCQ17 $-2e$; SCQ18 e^{1/z^2}; SCQ19′ $\frac{z^2+16}{(z^2-16)^2}$; SCQ20 $(1-t)e^{-t}$; SCQ21 -2; SCQ22 $\int_0^t f(t-s)g(s)ds$; SCQ05 -3; SCQ07 1st & 2nd; SCQ09 $0,1,-1$; Schröd. $b_n'=i(\pi^2n^2+1)b_n$; Wave-FT $\hat u_0\cos(\sqrt{\alpha^2+1}\,t)+\frac{\hat v_0}{\sqrt{\alpha^2+1}}\sin(\sqrt{\alpha^2+1}\,t)$; PDE $\sum e^{-\pi^2n^2t^2-t}g_n\sin(n\pi x)$.

EXAM-DAY CHECKLIST (avoid the classic traps)

1. **Poles:** cancel common factors *first* (a cancelled factor $\Rightarrow$ removable, not a pole).
2. **Residue ord.** m**:** multiply by $(z-z_0)^m$, differentiate $m-1$ times, divide by $(m-1)!$.
3. **Contour:** count only singularities *strictly inside*; "possible values" always include 0.
4. **Laplace:** never drop $-zy(0)-y'(0)$ in $\mathcal{L}[y'']$; complete the square for complex roots.
5. **PDE:** Dirichlet $\to$ sin; $u_{xxxx}\to+(\frac{n\pi}{L})^4$; a single $\sin(k\pi x)$ excites only $n=k$.
6. **Fourier:** $(i\alpha)^2=-\alpha^2$, $(i\alpha)^4=+\alpha^4$; heat/transport solution exponent is *negative*.
7. **Sanity:** a real integral must give a *real* number; recheck signs and $\frac{1}{i}=-i$.
8. **Marking:** unsure $\Rightarrow$ leave blank (wrong $=-\frac{1}{3}$ or -1).